AP Precalc Unit 2 Review Sheet: Exponential and Logarithmic Functions

A one-stop reference for tutoring Unit 2. The unit is about one idea: ***exponentials change by a constant factor, and logarithms undo them****.*

The unit at a glance

AP topic	What it covers	Anchor idea
2.1	Arithmetic vs geometric sequences	Add a constant vs multiply by a constant
2.2	Linear vs exponential change	Equal steps in x give equal *differences* (linear) or equal *ratios* (exponential)
2.3	Exponential functions	$f(x) = ab^x$: a is the start, b the growth factor
2.4	Exponential models in context	$b = 1 \pm r$ for a percent change r
2.5	Competing models	Random residuals mean a good fit; a pattern means the wrong model type
2.6	Composition	$f(g(x))$: inside first
2.7	Inverse functions	Swap x and y; reflect over $y = x$
2.8–2.9	Logarithms as inverses of exponentials	$\log_b x = y \iff b^y = x$
2.10	Logarithmic functions	Domain $x > 0$, vertical asymptote $x = 0$
2.11	Log properties	Product, quotient, power
2.12	Equations and inequalities	Same base, or take logs; check domain
2.13–2.14	Log modeling and semi-log plots	A straight line on a semi-log plot means exponential

1. Linear vs exponential (2.1–2.3)

- **Linear / arithmetic:** constant *difference*. $f(x) = mx + b$.
- **Exponential / geometric:** constant *ratio*. $f(x) = ab^x$, with $a = f(0)$.
- **Percent language:** $b = 1 + r$ for growth, $b = 1 - r$ for decay. A factor of 1.07 is 7% growth; 0.80 is 20% decay; 3 is 200% growth.
- **Features:** domain all reals; horizontal asymptote $y = 0$ (or $y = k$ for $ab^x + k$); $b > 1$ rises, $0 < b < 1$ falls; always concave up if $a > 0$.
- **Finding a and b from two points:** divide the equations to eliminate a. Through $(0, 6)$ and $(2, 54)$: $b^2 = 9$, $b = 3$.
- **Why the rate isn't constant:** the change over an interval is proportional to the current value, so the *average rate of change* grows (or shrinks) with x.

2. Modeling (2.4–2.5)

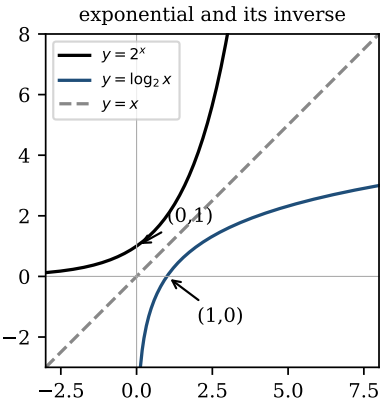
- **Model:** $A = a(1 \pm r)^t$. Say what t and A are, and give units.
- **Doubling / half-life:** use exponent t/period: $A(t) = A_0\left(\frac{1}{2}\right)^{t/h}$.
- **Competing models:** compare residual plots. **Random scatter around 0** means a good fit. A **U shape or other pattern** means the wrong function type.
- Residual = actual − predicted.

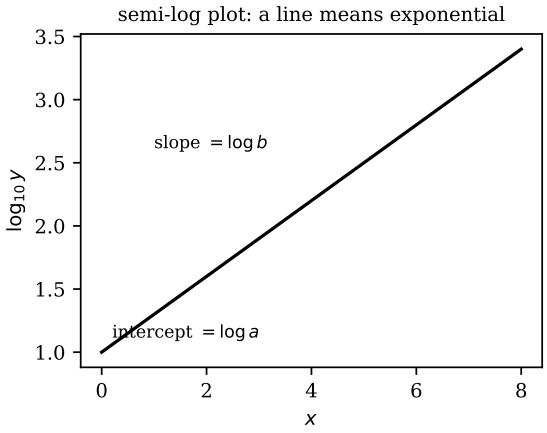
3. Composition and inverses (2.6–2.7)

- $(f \circ g)(x) = f(g(x))$: work from the inside out. Order matters.
- **Inverse:** swap x and y and solve. Domain and range of f become range and domain of f^{-1}.
- **One-to-one:** passes the horizontal line test. Only these have inverses (e.g. x^2 does not).
- Facts to say out loud: $f(f^{-1}(x)) = x$; if $f(a) = b$ then $f^{-1}(b) = a$; graphs reflect over $y = x$.

4. Logarithms (2.8–2.10)

- $\log_b x = y$ means $b^y = x$. Read it as **"what power of b gives x?"**
- $\log_b 1 = 0$, $\log_b b = 1$, $\log_b b^x = x$, $b^{\log_b x} = x$.
- log is base 10, ln is base e.
- **Graph of $\log_b x$:** domain $x > 0$; vertical asymptote $x = 0$; passes through $(1, 0)$ and $(b, 1)$; increasing if $b > 1$, always concave down.
- **Transformations:** in $a \log_b(x - h) + k$, the asymptote is $x = h$ and the domain is $x > h$. A negative a flips it over.





5. Properties (2.11)

Rule	Formula
Product	$\log_b(MN) = \log_b M + \log_b N$
Quotient	$\log_b(M/N) = \log_b M - \log_b N$
Power	$\log_b(M^p) = p \log_b M$
Change of base	$\log_b x = \dfrac{\ln x}{\ln b}$

There is no rule for $\log(M+N)$**.** This is the most common error.

6. Solving (2.12)

1. **Exponential equation, same base available:** write both sides with one base and set exponents equal. $4^{x+1} = 8^x \Rightarrow 2^{2x+2} = 2^{3x}$.
2. **Not the same base:** isolate the power, then take a log of both sides. $7^x = 20 \Rightarrow x = \ln 20 / \ln 7$.
3. **Log equation:** condense to one log, rewrite as an exponential, solve, then **check the domain**. Answers that make a log's input ≤ 0 are *extraneous*.
4. **Inequalities:** the direction flips only when you multiply or divide by a negative, or when the base is between 0 and 1.
5. **Quadratic form:** $2^{2x} - 5 \cdot 2^x + 4 = 0$ with $u = 2^x$ (stretch).

7. Log models and semi-log plots (2.13–2.14)

- **Log model:** $y = a + b \ln x$. Rises fast, then slows (concave down).
- **Semi-log plot:** plot $\log y$ against x. If it is a **straight line**, y is exponential: $\log y = \log a + x \log b$, so the slope is $\log b$ and the intercept is $\log a$.
- Useful for data spanning huge ranges (earthquakes, pH, decibels).

Common mistakes to watch for

- Writing $\log(a+b) = \log a + \log b$.
- Saying 0.85 is "85% decay" (it is 15% decay).
- Using $b = r$ instead of $b = 1 + r$.
- Forgetting to check log domain: extraneous solutions.
- Dropping the negative: $\log_{1/2} 8 = -3$, not 3.
- Treating f^{-1} as $1/f$.
- Using $\ln 0.5 / \ln 0.97$ with the wrong sign or swapped (check: a decay-to-half time must be positive).
- Mixing up $\ln(x^2)$ with $(\ln x)^2$.

Suggested session order (60 minutes)

1. 5 min: **retrieval warm-up.** What's the difference between linear and exponential?
2. 15 min: exponentials and models (2.2–2.4). Percent to factor.
3. 15 min: logs as inverses (2.8–2.10). Evaluate several by "what power?".
4. 15 min: properties and equations (2.11–2.12). Always check the domain.
5. 10 min: one modeling problem start to finish (half-life) and the semi-log idea.

Use the *In-Person Worksheet*: do the easy problem first; if she gets it, move to the hard one. The *Practice Test* is for homework or a timed run (about 45 minutes).